# GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

**Biology 2**
**0510**

**JUNE 2022** **ORDINARY LEVEL**

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| Subject Title | **Biology** |
| Paper No. | **Paper 2** |
| Subject Code No. | **0510** |

## Two and half hours

**Answer FIVE questions, ALL THREE (3) from SECTION A and any TWO (2) from SECTION B**

**All questions carry 20 marks each. For your guidance, the approximate mark for each part of a question is indicated in brackets.**

**Begin each question on a new page.**

**Give labelled diagrams where these will make the answers clearer.**

**You are advised to read carefully through the question paper before you begin your answers.**

**You are reminded of the necessity for good English and orderly presentation in your answers.**

**In calculations, you are advised to show all the steps in your working, giving your answer at each stage.**


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**SECTION B**
**Answer ALL 3 questions**

1. (a) Explain the following biological terms:
i) Osmosis.
ii) Diffusion.
iii) Active transport. **(4, 3, 2 marks)**
(b) How is osmosis important to plants? **(6 marks)**
(c) Describe an experiment to demonstrate osmosis in living tissues **(5 marks)**
**(Total = 20 marks)**

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2. (a) i) Briefly explain the term photosynthesis. **(5 marks)**
ii) Give the importance of photosynthesis in nature. **(3 marks)**
(b) How is the leaf suited for photosynthesis? **(12 marks)**
**(Total = 20 marks)**

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3. (a) Define the following terms:
i) Pollution.
ii) Producer. **(4, 5marks)**
(b) What appropriate measures can be put in place to control/prevent water pollution? **(6 marks)**
(c) Construct a food web using the following organisms: chicken, grasshopper, cabbage, hawk, goat, man. **(5 marks)**
**(Total = 20 marks)**

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**SECTION B**
**Answer any 2 questions**

4. (a) Define the term respiration. . **(5 marks)**
(b) Compare aerobic and anaerobic respiration **(6 marks)**
(c) Describe an experiment to show that energy is released during respiration. (**9 marks)**
**(Total = 20 marks)**

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5. (a) Write short notes on the following biological terms:
i) Stimulus.
ii) Tropism.
iii) Reflex action.
iv) Effector. **(4, 4, 4, 2 marks)**
(b) With the aid of a diagram, describe the effect of illuminating a shoot from one direction. **(6 marks)**
**(Total = 20 marks)**

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6. (a) Define the following terms:
i) Copulation.
ii) Ovulation.
iii) Fertilization. **(3, 3 ,4 marks)**
(b) Describe the process of birth in humans. **(8 marks)**
(c) State two functions of the placenta. **(2 marks)**
**(Total =20 marks)**

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7. (a) What is:
i) Genetics,
ii) Variation,
iii) Mutation? **(2, 3, 3 marks)**
(b) When a red-flowered plant was crossed with a white flowered plant of the same species, all the resulting offspring of the F1 generation were red flowered plants. When one of the F1 offspring was self-pollinated, 176 red-flowered and 64 white-flowered plants of the F2 generation were obtained.
i) Explain the genetics of these crosses by using appropriate diagrams and symbols. **(10 marks)**
ii) What is the importance of a test cross? **(2 marks)**
**(Total =20 marks)**

**GO BACK AND CHECK YOUR WORK**

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